

**OBJECTIVE-**The main purpose of this lab is to give the student an opportunity to learn how to build an HMI application. This lab uses the Allen Bradley HMI platform.

**PROCEDURE-**The main procedure in this lab is to first write the code in plain ladder logic, and then link that code to an HMI panel. This lab is a slight variation of the earlier cash register lab from chapter 7. In this lab, there are two screens. The first screen displays the food items and the quantity ordered. There are buttons for placing the customer's order. If the customer wishes to dine in the store, then a 6.25% sales tax is applied. Otherwise, if the order is to go, then no tax is applied. The appropriate food bill is displayed on the first screen. If the manager wants to change the price of food items, then there is a button to go to a second screen where food prices can be entered into input fields. On the second screen, there is another button to return to the first screen. The original wired lab 7.2 was a rather straightforward lab where everything was displayed directly in the program window, and all the inputs came from buttons on a breadboard. But in this lab, all the inputs and outputs are linked to HMI elements such as buttons and text I/O fields. The code works through a process called OLE, Object Linking and Embedding which is an old concept dating back to the early days of MS Visual Basic.

**LEARNING OUTCOME-** This project taught the student how to build an effective HMI application using the Allen Bradley platform. The software used for this was Studio 5000 Visual Designer. Doing this lab was a very enjoyable experience for the student because the process of building an HMI screen is closer to a high-level programming experience. This lab also gave the student a chance to see how powerful HMI programming can be since through HMI, it is entirely possible to build some very powerful applications on a plc and so extend the capabilities of what a plc program can do.